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Real Effects of the Sovereign Debt Crisis in Europe: Evidence from Syndicated Loans

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1 Summary

This paper studies how the European sovereign debt crisis affected bank lending and the real activity of nonfinancial firms. The central idea is that the crisis impaired banks directly through losses on sovereign bond holdings and indirectly through incentives for weak banks to shift risk into distressed sovereign debt. These banking-sector responses reduced loan supply, and firms connected to the affected banks cut investment, employment, and sales.

The paper separates three channels. The **balance-sheet-hit channel** operates because declining GIIPS sovereign bond prices reduced bank equity and forced affected banks to deleverage. The **risk-shifting channel** operates because weakly capitalized banks with large preexisting exposure to risky sovereigns had incentives to buy even more of those sovereign bonds, crowding out corporate lending. The **moral-suasion channel** would operate if distressed sovereign governments pressured domestic banks to buy domestic government debt. The paper finds strong evidence for the first two channels and little evidence that moral suasion was the key driver.

The empirical design uses detailed bank-firm relationships from DealScan, firm characteristics from Amadeus, and bank-level sovereign debt holdings from the EBA stress tests, capital exercises, annual reports, and quarterly reports. The paper first studies how banks changed sovereign holdings and loan supply during 2010–2012. It then connects firms to their lenders' sovereign risk exposures and studies whether exposed firms experienced worse real outcomes.

The main banking result is that crisis-exposed and weak banks reduced credit supply substantially. For non-GIIPS banks, the banks most prone to risk shifting increased their highest preexisting GIIPS sovereign exposure by roughly 30 percent and reduced corporate lending. For GIIPS banks, weak banks with strong risk-shifting incentives increased risky sovereign exposure even more, and their borrowers became significantly less likely to obtain a new syndicated loan. The paper does not find that weak government-controlled GIIPS banks, the banks most likely to face moral suasion, were the main source of reduced lending.

The main real-effect result is that firms indirectly exposed to GIIPS sovereign risk through their banks suffered economically large declines in activity. Firms connected to more affected banks had lower employment growth, lower investment, and lower sales growth than comparable firms connected to less affected banks. The paper estimates that the credit crunch explains a large share of the total real effects suffered by European firms during the crisis: between 44 percent and 66 percent depending on the country group and outcome.

2 Big picture

- **Question.** Did the European sovereign debt crisis reduce bank loan supply, and did this credit contraction depress firm real activity?
- **Main answer.** Yes. Sovereign exposures impaired banks and induced risk shifting, which reduced syndicated lending and caused affected firms to cut investment, employment, and sales.
- **Key tension.** GIIPS sovereign debt purchases by banks could reflect risk shifting, government pressure, or balance-sheet losses. The paper disentangles these channels using bank type, bank nationality, hedging behavior, and preexisting sovereign exposure.
- **Identification intuition.** The authors use bank-firm relationships to compare firms with similar demand conditions but different exposure to banks hit by the sovereign crisis.
- **Main contribution.** The paper links sovereign debt stress to bank portfolio choices, loan supply, and firm-level real outcomes in one empirical framework.

3 Institutional setting and mechanisms

3.1 Sovereign stress and bank balance sheets

The sovereign crisis began when Greece, Ireland, Italy, Portugal, and Spain faced rising refinancing concerns. Because European banks held large quantities of sovereign debt, sovereign stress fed directly into bank balance sheets. A decline in the value of preexisting sovereign holdings reduced bank equity and tightened lending capacity.

3.2 Balance-sheet-hit channel

A bank with large holdings of GIIPS sovereign debt suffered losses when those bonds declined in value. This balance-sheet impairment reduced the bank's ability or willingness to extend new loans. This channel can affect both GIIPS and non-GIIPS banks, as long as they held risky sovereign debt before the crisis.

3.3 Risk-shifting channel

A weak bank with existing exposure to a risky sovereign may rationally increase its exposure if downside losses are partly borne by creditors, taxpayers, or rescue mechanisms. This risk-shifting logic predicts that weak banks should increase the sovereign asset to which they already have the highest risky exposure, and that such purchases should crowd out corporate lending.

3.4 Moral-suasion channel

Moral suasion predicts that GIIPS governments pressured domestic banks, especially government-controlled banks, to buy domestic sovereign debt. Unlike risk shifting, moral suasion should be most visible among domestic government-controlled banks and may be accompanied by hedging if banks dislike the extra exposure.

4 Data and key variables

4.1 Bank-firm links

The paper uses Thomson Reuters LPC's DealScan to identify syndicated loan relationships. The empirical design uses these precrisis bank-firm links to measure whether a firm is exposed to a lender affected by the sovereign crisis.

4.2 Firm data

Firm-level balance sheet and income statement variables come from Bureau van Dijk's Amadeus database. Key outcomes include changes in cash holdings, net debt, interest coverage, credit lines, undrawn credit lines, employment growth, investment, and sales growth.

4.3 Bank data

The bank-side data include sovereign debt holdings from EBA stress tests and capital exercises, plus annual and quarterly reports. The paper also uses ratings, CDS spreads, bank balance sheet variables, and ownership information to classify banks as weak, high-exposure, or government-controlled.

4.4 Core exposure measures

- **Net Change Risk-Shifting Asset.** Increase in net holdings of the GIIPS sovereign debt to which a bank had the highest preexisting exposure.
- **Gross Change Risk-Shifting Asset.** Increase in gross holdings of the same risky sovereign debt.
- **Value Change GIIPS Sovereign Debt.** Bank-level value loss on preexisting GIIPS sovereign debt holdings during the crisis.
- **Indirect GIIPS Sovereign Risk.** Firm-level exposure to GIIPS sovereign risk through the firm's bank lending relationships.

5 Empirical strategy

5.1 Bank-level tests

The first step studies banks' sovereign debt portfolios. For non-GIIPS banks, the key test is whether weak banks with high preexisting exposure increased the risky GIIPS sovereign asset to which they were already exposed. Since non-GIIPS banks are not directly subject to GIIPS government pressure, this helps isolate risk shifting.

For GIIPS banks, the paper compares predictions of risk shifting and moral suasion. Risk shifting predicts that banks increase the risky asset to which they already have high exposure and do not hedge the new exposure. Moral suasion predicts stronger domestic sovereign purchases by government-controlled domestic banks and potentially hedging of unwanted exposure.

5.2 Loan-supply tests

The paper uses modified Khwaja and Mian-style regressions to test whether affected banks reduced lending to existing borrowers. The unit is a bank-year-firm cluster. Borrower fixed effects or borrower-year-type controls help distinguish loan supply from loan demand.

5.3 Firm real-effect tests

The firm-level analysis relates firms' real outcomes to their indirect exposure to GIIPS sovereign risk through their banks. The paper controls for firm fixed effects and rich industry-country-time or country-year fixed effects. The logic is that otherwise similar firms should differ in real outcomes if one group had precrisis relationships with more crisis-exposed banks.

5.4 Demand-side and macro shock checks

The paper addresses alternative explanations in two main ways. First, it studies non-GIIPS firms without direct business exposure to GIIPS countries, including firms whose GIIPS bank links arose through pre-sample takeovers. Second, it shows that the effects are stronger for firms less able to substitute bank credit, such as private, unrated, or bond-market-inactive firms.

6 Main results

6.1 Evidence for risk shifting among non-GIIPS banks

Non-GIIPS banks with weak balance sheets and high preexisting GIIPS exposure increased the sovereign debt to which they were most exposed. This behavior is difficult to explain by moral suasion because these banks were outside GIIPS countries. The increase in risky sovereign exposure crowded out corporate lending.

6.2 GIIPS banks: risk shifting versus moral suasion

For GIIPS banks, the evidence points more toward risk shifting than moral suasion. Weak banks with strong risk-shifting incentives increased their highest preexisting GIIPS sovereign exposure more than other banks and did not hedge the additional exposure. In contrast, banks most likely to face moral suasion, such as weak government-controlled banks, were not the main source of the lending contraction.

6.3 Loan supply contracts

Borrowers connected to affected banks became less likely to obtain new syndicated loans. The results are strongest for banks suffering balance-sheet losses and for banks increasing risky sovereign exposures. The paper interprets this as direct evidence of a bank lending channel from sovereign stress to corporate credit.

6.4 Real effects on firms

Firms with higher indirect exposure to GIIPS sovereign risk through their banks experienced lower employment growth, lower investment, and lower sales growth. The real effects are economically meaningful: the paper reports that exposed firms recorded 4.7 percentage points lower employment growth, 5.9 percentage points lower investment, and 5.6 percentage points lower sales growth relative to comparable firms.

6.5 Importance for aggregate real effects

The estimated credit crunch explains a large part of the crisis-period contraction in firm outcomes. For GIIPS countries, the credit crunch explains about 61 percent of the sales decrease, 62 percent of the investment contraction, and 66 percent of the employment decline. For non-GIIPS European countries, the corresponding shares are about 44 percent, 46 percent, and 49 percent.

7 Interpretation and intuition

- **Sovereign-bank nexus.** Sovereign stress is not only a government-finance problem; it can directly impair bank balance sheets and corporate lending.
- **Risk shifting matters.** Weak banks may double down on risky sovereign assets instead of reducing risk, especially when downside losses are partly externalized.
- **Moral suasion is not enough to explain the data.** Domestic sovereign purchases occurred, but the patterns by bank ownership, hedging, and preexisting exposure are more consistent with risk shifting.

- **Bank relationships transmit shocks.** Firms are affected through their lender network even when the firms themselves are not directly exposed to sovereign stress.
- **Real outcomes depend on financing substitution.** Firms that cannot easily replace bank credit suffer larger real effects.

8 Caveats and discussion

- **Syndicated-loan focus.** The analysis is based on the syndicated loan market, which is important for European firms but does not cover every form of bank credit.
- **Exposure measurement.** Firm exposure is measured through preexisting lending relationships and banks' sovereign portfolios; this is powerful but not a direct observation of every credit negotiation.
- **Moral suasion may still exist.** The evidence suggests it was not the main driver of the documented lending contraction, not that political pressure was completely absent.
- **External validity.** The setting is the European sovereign debt crisis, where bank sovereign exposures and the eurozone institutional environment are central.

9 Notes-created summary tables

9.1 Research design at a glance

Element	Implementation	Purpose
Shock	European sovereign debt crisis, especially GIIPS sovereign stress.	Creates bank-level variation in sovereign losses and risky sovereign holdings.
Bank-side data	DealScan bank-firm links; EBA and bank reports for sovereign holdings; bank ratings and CDS spreads.	Measures bank exposure, weakness, and lending changes.
Firm-side data	Amadeus firm balance sheets and outcomes.	Measures real effects on investment, employment, and sales.
Loan-supply design	Modified Khwaja and Mian regressions using bank-firm relationships.	Separates bank credit supply from borrower demand.
Firm real effects	Regress firm outcomes on indirect sovereign exposure through lenders.	Shows how bank distress transmits to nonfinancial firms.
Main channel tests	Compare balance-sheet hit, risk shifting, and moral suasion predictions.	Identifies which mechanisms explain the credit crunch.

9.2 Mechanisms

Channel	Prediction	Evidence in paper
Balance-sheet hit	Banks with larger value losses on preexisting GIIPS sovereign holdings reduce lending.	Supported; impaired banks are less likely to extend new loans.
Risk shifting	Weak banks with high preexisting risky sovereign exposure buy more of that risky sovereign debt and reduce lending.	Strongly supported, for both non-GIIPS and GIIPS banks.
Moral suasion	Domestic government-controlled GIIPS banks are pressured to buy domestic sovereign debt and reduce lending.	Not the main driver; the strongest patterns are inconsistent with this being dominant.
Credit crunch	Borrowers linked to affected banks receive less credit and cut real activity.	Supported by loan-level and firm-level outcomes.

9.3 Main empirical takeaways

Result	Interpretation
Affected banks reduced the probability of borrowers obtaining new syndicated loans.	Sovereign stress transmitted into corporate credit supply.
Non-GIIPS weak high-exposure banks increased risky GIIPS sovereign holdings.	Evidence for risk shifting without relying on domestic government pressure.
GIIPS weak high-exposure banks increased risky sovereign exposure without hedging.	Further evidence for risk shifting.
Government-controlled weak GIIPS banks were not the main contraction source.	Moral suasion was not the central mechanism in the main results.
Exposed firms reduced employment, investment, and sales growth.	The bank credit contraction had real consequences.
Effects are larger for private, unrated, or bond-market-inactive firms.	Firms that cannot substitute financing suffer more.

10 Original figures and tables

Table 1

Variable definitions

Variable	Definition
Bank level: Affected bank measures based on sovereign debt holdings data	
<i>Net Change Risk-Shifting Asset_t</i>	Increase in net holdings of the GIIPS sovereign debt to which a bank had the highest preexisting exposure
<i>Gross Change Risk-Shifting Asset_t</i>	Increase in gross holdings of the GIIPS sovereign debt to which a bank had the highest preexisting exposure
<i>Net Change All GIIPS_t</i>	Increase in net holdings of a bank's total GIIPS sovereign debt
<i>Gross Change All GIIPS_t</i>	Increase in gross holdings of a bank's total GIIPS sovereign debt
<i>Crisis</i>	Dummy equal to one if observation falls into sovereign debt crisis period of 2010 to 2012
<i>Value Change GIIPS Sov Debt 2010_t</i>	Change in value of the bank's preexisting (i.e., March 2010) sovereign debt holdings normalized by equity
<i>Total Value Change GIIPS Sov Debt 2010</i>	Sum of <i>Value Change GIIPS Sov Debt 2010_t</i> over the crisis period
<i>Risk Change GIIPS Sov Debt 2010_t</i>	Change in risk of the bank's preexisting (i.e., March 2010) sovereign debt holdings normalized by equity
<i>Total Risk Change GIIPS Sov Debt 2010</i>	Sum of <i>Risk Change GIIPS Sov Debt 2010_t</i> over the crisis period
<i>No Hedge</i>	Dummy equal to one if a bank increases gross and net holdings by at least the same amount (i.e., bank is not hedging)
<i>GIIPS Sov Risk</i>	Sum of fraction of GIIPS sovereign debt holdings weighted by the countries CDS over total assets
Bank level: Affected bank measures based on ex ante bank-level characteristics	
<i>Low-Quality</i>	Dummy equal to one if a bank has an above-median leverage ratio or a below-median rating in 2009
<i>High-Exposure</i>	Dummy equal to one if a bank's risk-shifting asset exposure is above median in March 2010 (separate medians for GIIPS/non-GIIPS banks)
<i>Gov-Control</i>	Dummy equal to one if a 2009 bank has positive government ownership or if its fraction of government-affiliated directors is above median
Firm level: Key explanatory variables	
<i>Indirect GIIPS Sov Risk</i>	Measures the GIIPS sovereign risk each firm is exposed to via its bank lending relationships
<i>High (Low) Indirect GIIPS Sov Risk</i>	Indicator variable equal to one if a firm's <i>Indirect GIIPS Sov Risk</i> is above (below) the sample median
Firm level: Dependent variables (winsorized at the 5% level)	
<i>Net Debt</i>	$\frac{Current + Non-Current Liabilities - Cash}{Total Assets}$
$\Delta Cash$	$\frac{Cash_{t+1} - Cash_t}{Total Assets_t}$
<i>Interest Coverage Ratio (ICR)</i>	$\frac{EBIT}{Interest Expense}$
<i>Employment Growth (Emp Growth)</i>	$\ln(Employment_{t+1}) - \ln(Employment_t)$
<i>CAPX</i>	$\frac{Fixed Assets_{t+1} - Fixed Assets_t + Depreciation}{Fixed Assets_t}$, set to 0 if negative
<i>Sales Growth</i>	$\ln(Sales_{t+1}) - \ln(Sales_t)$
Firm level: Control variables (winsorized at the 5% level)	
<i>ln(Assets)</i>	Natural logarithm of total assets
<i>Leverage</i>	$\frac{Total Assets - Total Equity}{Total Assets}$
<i>Net Worth</i>	$\frac{Total shareholder funds \& Liabilities - Current \& Non-Current Liabilities - Cash}{Total Assets}$
<i>Tangibility</i>	$\frac{Fixed Assets}{Total Assets}$
<i>EBITDA/Assets</i>	$\frac{EBITDA}{Total Assets}$
<i>Cash Flow</i>	$\frac{Cash Flow}{Total Assets}$

Table 1. Variable definitions

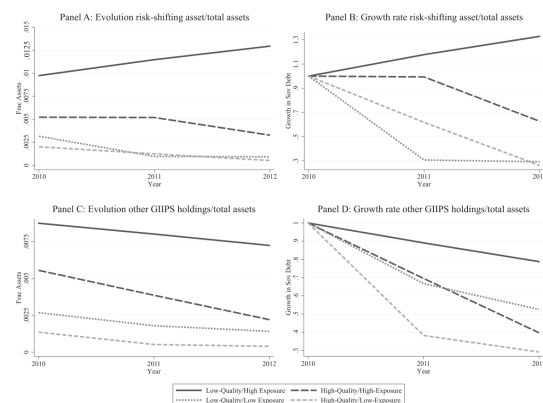


Figure 1

GIIPS sovereign debt holdings: Non-GIIPS banks

Panel A shows the evolution of the net risk-shifting asset holdings over total assets for non-GIIPS banks over the crisis period 2010 to 2012 (the growth rate is depicted in panel B). The risk-shifting asset is defined as the GIIPS sovereign debt to which a bank had the highest exposure at the beginning of the crisis. Panel C shows the evolution of the average of the remaining four GIIPS sovereign debt holdings (growth rate depicted in panel D). A non-GIIPS bank is classified as low quality if it has an above-median leverage ratio or a below-median rating in 2009. A non-GIIPS bank is classified as high exposure if its risk-shifting asset exposure as of March 2010, the date of the first EBA stress test, is above the sample median for non-GIIPS banks.

Figure 1. GIIPS sovereign debt holdings: Non-GIIPS banks

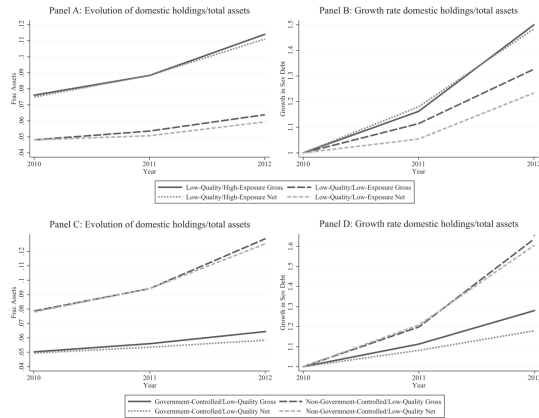


Figure 2. Domestic sovereign debt holdings: Low-quality GIIPS banks
Panels A and B show the evolution of gross and net domestic sovereign debt holdings over total assets for low-quality GIIPS banks over the crisis period 2010 to 2012 (the growth rate is depicted in panels B and D). A GIIPS bank is classified as low quality if it has an above-median leverage ratio or a below-median rating in 2009. A GIIPS bank is classified as high exposure if its risk-shifting asset exposure as of March 2010, the date of the first EBA stress test, is above the sample median for GIIPS banks. We classify a GIIPS bank as being government controlled if it has either positive government ownership or an above-median fraction of government-affiliated directors in 2009.

Figure 2. Domestic sovereign debt holdings: Low-quality GIIPS banks

Table 3
Bank lending regressions (sovereign debt holdings data): GIIPS banks

Panel A: Change in loan volume						
	(1) Δ Volume	(2) Δ Volume	(3) Δ Volume	(4) Δ Volume	(5) Δ Volume	(6) Δ Volume
Gross Change Risk-Shifting Asset	-1.010*** (-3.44)			-0.740** (-2.04)		0.421 (1.01)
Net Change Risk-Shifting Asset		-1.501*** (-3.40)		-1.278** (-2.65)		
Value Change GIIPS Sov Debt 2010			0.152** (2.76)	0.121** (2.09)	0.115** (2.00)	0.121** (2.81)
Gross Change Risk-Shifting Asset $\times$ No Hedge					-1.730** (-2.24)	
R^2	0.822	0.812	0.843	0.845	0.844	0.847
N	1,514	1,514	1,514	1,514	1,514	1,514
Panel B: Loan decrease						
	Loan Decrease	Loan Decrease	Loan Decrease	Loan Decrease	Loan Decrease	Loan Decrease
Gross Change Risk-Shifting Asset	0.012** (2.25)			0.011** (2.15)		-0.007 (-1.19)
Net Change Risk-Shifting Asset		0.019*** (3.36)			0.017** (2.36)	
Value Change GIIPS Sov Debt 2010			-0.002** (-2.20)	-0.002** (-2.21)	-0.002** (-1.97)	-0.003** (-2.60)
Gross Change Risk-Shifting Asset $\times$ No Hedge						0.036** (2.70)
R^2	0.765	0.766	0.779	0.795	0.780	0.814
N	1,514	1,514	1,514	1,514	1,514	1,514
Panel C: Change in loan spread						
	Δ Spread	Δ Spread	Δ Spread	Δ Spread	Δ Spread	Δ Spread
Gross Change Risk-Shifting Asset	4.268** (2.20)			5.210** (2.08)		-2.894 (-1.54)
Net Change Risk-Shifting Asset		6.195** (2.24)			6.568* (2.04)	
Value Change GIIPS Sov Debt 2010			-0.615*** (-2.98)	-0.529* (-1.81)	-0.517* (-1.87)	-0.541*** (-3.04)
Gross Change Risk-Shifting Asset $\times$ No Hedge						6.900** (2.33)
R^2	0.842	0.832	0.815	0.840	0.835	0.867
N	769	769	769	769	769	769
Bank Level Controls	YES	YES	YES	YES	YES	YES
Firm Cluster-Bank Fixed Effects	YES	YES	YES	YES	YES	YES
Firm Cluster-Time Fixed Effects	YES	YES	YES	YES	YES	YES

Table 3. Bank lending regressions: GIIPS banks

Table 2
Bank lending regressions (sovereign debt holdings data): Non-GIIPS banks

Panel A: Change in loan volume					
	(1) Δ Volume	(2) Δ Volume	(3) Δ Volume	(4) Δ Volume	(5) Δ Volume
Net Change Risk-Shifting Asset	-2.683** (-2.75)		-3.011*** (-3.47)		
Gross Change Risk-Shifting Asset				-2.647** (-2.75)	-2.971*** (-3.47)
Value Change GIIPS Sov Debt 2010		0.329** (2.16)	0.332** (2.28)		0.332** (2.28)
R^2	0.779	0.795	0.800	0.779	0.800
N	2,384	2,384	2,384	2,384	2,384
Panel B: Loan decrease					
	Loan Decrease	Loan Decrease	Loan Decrease	Loan Decrease	Loan Decrease
Net Change Risk-Shifting Asset	0.040** (2.38)		0.044** (2.37)		
Gross Change Risk-Shifting Asset				0.039** (2.38)	0.043** (2.37)
Value Change GIIPS Sov Debt 2010		-0.004** (-2.08)	-0.005** (-2.60)		-0.005** (-2.60)
R^2	0.792	0.794	0.803	0.792	0.803
N	2,384	2,384	2,384	2,384	2,384
Panel C: Change in loan spread					
	Δ Spread	Δ Spread	Δ Spread	Δ Spread	Δ Spread
Net Change Risk-Shifting Asset	10.762* (1.71)		10.888*** (3.81)		
Gross Change Risk-Shifting Asset				10.972* (1.71)	11.098*** (3.81)
Value Change GIIPS Sov Debt 2010		-1.145** (-2.02)	-1.289*** (-3.24)		-1.289*** (-3.24)
R^2	0.784	0.734	0.826	0.784	0.826
N	1,419	1,419	1,419	1,419	1,419
Bank Level Controls	YES	YES	YES	YES	YES
Firm Cluster-Bank Fixed Effects	YES	YES	YES	YES	YES
Firm Cluster-Time Fixed Effects	YES	YES	YES	YES	YES

Table 2. Bank lending regressions: Non-GIIPS banks

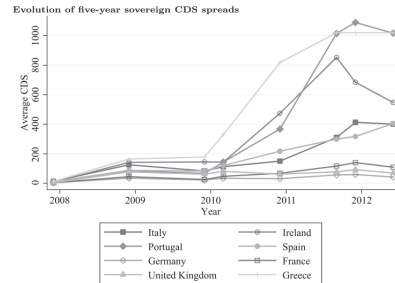


Figure 3
Evolution of five-year sovereign CDS spreads
This figure plots the evolution of the five-year sovereign CDS spreads for European countries.

Figure 3. Evolution of five-year sovereign CDS spreads

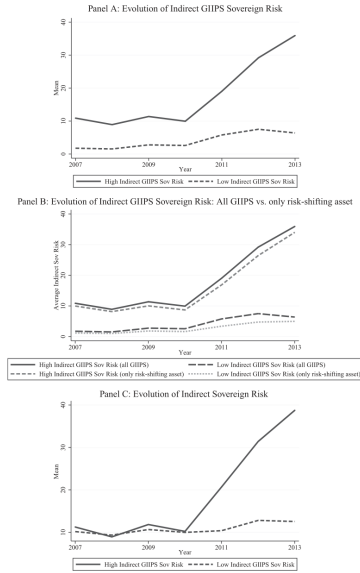


Figure 4
Evolution of indirect sovereign risk.
This figure shows the evolution of the firms' indirect exposure to sovereign risk. Panel A shows the evolution of the firms' *Indirect GIIPS Sovereign Risk*, which measures the GIIPS sovereign risk each firm is exposed to via its bank lending relationships. Panel B shows in addition the evolution of only the indirect exposure to the risk-shifting asset, and panel C the evolution of the firms' indirect exposure to all sovereign debt. The risk-shifting asset is defined as the GIIPS sovereign debt to which a bank had the highest exposure at the beginning of the crisis. In all panels we split firms into high (i.e., above the precrisis median) and low (i.e., below the median) *Indirect GIIPS Sov Risk* firms.

Figure 4. Evolution of indirect sovereign risk

Table 4
Real effects: Entire sample

	(1) Δ Cash	(2) Net Debt	(3) ICR	(4) Credit Line	(5) Undrawn	(6) Emp Growth	(7) CAPX	(8) Sales Growth
Indirect GIIPS Sov Risk	-0.000 (-0.74)	-0.008** (-2.50)	-0.656** (-2.16)	-0.012** (-2.10)	-0.016** (-2.27)	-0.007** (-2.17)	-0.009** (-2.12)	-0.008** (-2.34)
Indirect GIIPS Sov Risk $\times$ Cashflow	0.007*** (3.15)							
R^2	0.462	0.856	0.833	0.876	0.813	0.610	0.716	0.610
N	3,853	4,272	4,318	491	491	3,630	4,280	4,055
Firm Controls	YES	YES	YES	YES	YES	YES	YES	YES
Firm Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Country-Year Fixed Effects	NO	NO	NO	YES	YES	NO	NO	NO
Industry-Country-Year Fixed Effects	YES	YES	YES	NO	NO	YES	YES	YES
Foreign Bank GIIPS Country-Year Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES

Table 4. Real effects: Entire sample

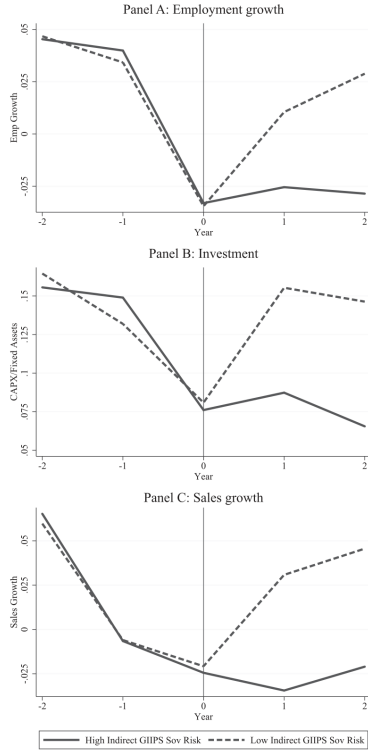


Figure 5
Evolution of real outcome variables for entire firm sample
This figure plots the evolution of employment growth, investment, and sales growth in the precrisis period (years -2 and -1) and the crisis period (starting in year 0) for the full sample of firms, separately for high (i.e., above the precrisis median) and low (i.e., below the median) *Indirect GIIPS Sov Risk* firms. *Indirect GIIPS Sov Risk* measures the GIIPS sovereign risk each firm is exposed to via its bank lending relationships.

Figure 5. Evolution of real outcome variables for entire sample

Table 5
Descriptive statistics precrisis

	Panel A: Dependent variables			Panel B: Explanatory variables						
	Emp. Growth	CAPX	Sales Growth	Total Assets (mm)	Tangibility	KR	Net Worth	EBITDA/Assets	Leverage	
High Indirect GIIPS Sov Risk										
Mean	0.054	0.195	0.055	3640	0.572	2.997	0.223	0.110	0.617	
Median	0.032	0.115	0.055	372	0.611	1.440	0.392	0.097	0.650	
Std. Dev.	0.156	0.243	0.217	7510	0.263	4.100	0.179	0.103	0.181	
Low Indirect GIIPS Sov Risk										
Mean	0.045	0.192	0.048	3220	0.551	3.009	0.219	0.117	0.606	
Median	0.022	0.113	0.053	565	0.574	2.020	0.252	0.105	0.577	
Std. Dev.	0.156	0.246	0.199	6870	0.264	3.359	0.175	0.102	0.219	
Diff. (t-stat)	0.009 (1.10)	0.003 (0.27)	0.007 (0.69)	420 (1.32)	0.021 (1.84)	-0.415 (-0.07)	0.482 (0.01)	-0.007 (-1.52)	0.011 (1.27)	
Correlation with Indirect GIIPS Sov Risk					-0.02	0.04	-0.08	-0.03	-0.06	0.07

Table 5. Descriptive statistics precrisis

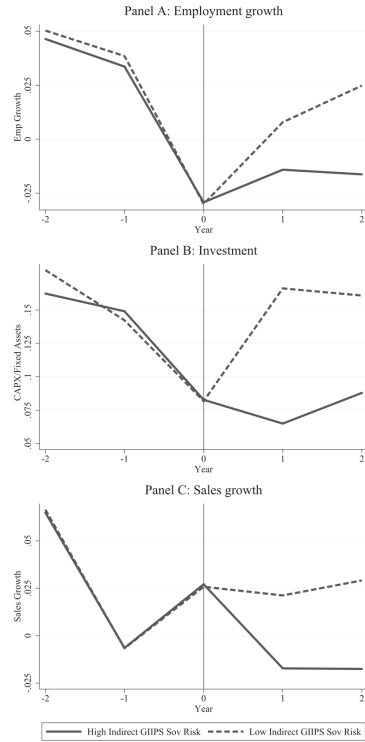


Figure 6
Evolution of real outcome variables for non-GIIPS firms without GIIPS or other non-EU subsidiaries
This figure plots the evolution of employment growth, investment, and sales growth in the precrisis period (years -2 and -1) and the crisis period (starting in year 0) for the sample of non-GIIPS firms without GIIPS or other non-EU subsidiaries, separately for high (i.e., above the precrisis median) and low (i.e., below the median) *Indirect GIIPS Sov Risk* firms. *Indirect GIIPS Sov Risk* measures the GIIPS sovereign risk each firm is exposed to via its bank lending relationships.

Figure 6. Evolution of real outcome variables for non-GIIPS firms

Table 6

Real effects: Subsidiaries

	Δ Cash	Net Debt	ICR	Emp Growth	CAPX	Sales Growth
Panel A: Non-GIIPS firms without GIIPS or other non-EU subsidiaries						
Indirect GIIPS Sov Risk	-0.001 (-0.69)	-0.005** (-2.38)	-0.407** (-2.55)	-0.005** (-2.47)	-0.009** (-2.46)	-0.008** (-2.29)
Indirect GIIPS Sov Risk $\times$ Cashflow	0.006*** (3.83)					
R^2	0.629	0.900	0.922	0.756	0.801	0.671
N	934	1,098	1,192	832	1,039	1,021
Panel B: Non-GIIPS firms without GIIPS subsidiaries (M&A only)						
Indirect GIIPS Sov Risk	-0.002 (-1.21)	-0.007** (-2.60)	-0.393** (-2.47)	-0.006** (-2.14)	-0.008** (-2.38)	-0.008** (-2.14)
Indirect GIIPS Sov Risk $\times$ Cashflow	0.006** (2.46)					
R^2	0.598	0.896	0.926	0.735	0.789	0.666
N	832	999	1,056	746	931	909
Panel C: Non-GIIPS firms without GIIPS subsidiaries (nontradable sectors)						
Indirect GIIPS Sov Risk	-0.001 (-0.81)	-0.007*** (-3.01)	-0.462** (-2.51)	-0.006** (-2.29)	-0.007** (-2.41)	-0.008** (-2.52)
Indirect GIIPS Sov Risk $\times$ Cashflow	0.007*** (5.12)					
R^2	0.633	0.909	0.953	0.731	0.804	0.697
N	811	950	1,012	693	884	863
Panel D: Non-GIIPS firms with GIIPS subsidiaries						
Indirect GIIPS Sov Risk	-0.001 (-0.69)	0.000 (0.00)	-0.206 (-0.55)	0.001 (0.72)	0.002 (1.61)	-0.002 (-0.73)
Indirect GIIPS Sov Risk $\times$ Cashflow	0.002 (0.47)					
R^2	0.511	0.866	0.750	0.584	0.705	0.575
N	1,242	1,274	1,291	1,151	1,263	1,261
Firm Controls	YES	YES	YES	YES	YES	YES
Firm Fixed Effects	YES	YES	YES	YES	YES	YES
Industry-Time Fixed Effects	YES	YES	YES	YES	YES	YES
Foreign GIIPS Bank-Country-Time Fixed Effects	YES	YES	YES	YES	YES	YES

Table 6. Real effects: Subsidiaries

Table 7
Real effects: Firms' ability to substitute loan supply decrease

	Δ Cash	Net Debt	ICR	Emp Growth	CAPX	Sales Growth
Panel A: Listed vs. unlisted firms						
Indirect GIIPS Sov Risk	0.000 (0.62)	0.000 (0.63)	-0.100 (-0.88)	-0.000 (-0.04)	-0.000 (-0.08)	-0.001 (-0.21)
Indirect GIIPS Sov Risk $\times$ Cashflow	-0.002 (-1.22)					
Indirect GIIPS Sov Risk $\times$ Private	-0.001 (-0.91)	-0.011** (-2.15)	-0.767** (-2.51)	-0.009** (-2.26)	-0.011** (-2.51)	-0.011*** (-2.63)
Indirect GIIPS Sov Risk $\times$ Private $\times$ Cashflow	0.010** (2.16)					
R^2	0.467	0.849	0.847	0.610	0.727	0.603
N	3,853	4,272	4,518	3,630	4,250	4,055
Panel B: Rated vs. unrated firms						
Indirect GIIPS Sov Risk	0.000 (0.54)	0.002 (1.30)	0.020 (0.18)	0.000 (0.55)	0.001 (1.17)	0.000 (0.70)
Indirect GIIPS Sov Risk $\times$ Cashflow	0.001 (0.28)					
Indirect GIIPS Sov Risk $\times$ No Rating	-0.001 (-1.17)	-0.012** (-2.06)	-0.856*** (-3.01)	-0.010*** (-2.90)	-0.012*** (-3.07)	-0.012*** (-3.00)
Indirect GIIPS Sov Risk $\times$ No Rating $\times$ Cashflow	0.009** (2.05)					
R^2	0.467	0.858	0.851	0.612	0.738	0.604
N	3,853	4,272	4,518	3,630	4,250	4,055
Panel C: Firms with vs. without bond issues						
Indirect GIIPS Sov Risk	0.000 (0.30)	0.002 (1.23)	-0.043 (-0.39)	0.000 (0.11)	0.001 (1.32)	0.000 (0.30)
Indirect GIIPS Sov Risk $\times$ Cashflow	-0.002 (-0.84)					
Indirect GIIPS Sov Risk $\times$ No Bond Outstanding	-0.001 (-0.92)	-0.012** (-2.12)	-0.801*** (-2.59)	-0.009** (-2.29)	-0.012*** (-3.13)	-0.011** (-2.05)
Indirect GIIPS Sov Risk $\times$ No Bond Outstanding $\times$ Cash Flow	0.008** (2.04)					
R^2	0.477	0.858	0.851	0.610	0.735	0.604
N	3,853	4,272	4,518	3,630	4,250	4,055
Firm Controls	YES	YES	YES	YES	YES	YES
Firm Fixed Effects	YES	YES	YES	YES	YES	YES
Industry-Country-Time Fixed Effects	YES	YES	YES	YES	YES	YES
Foreign Bank GIIPS Country-Time Fixed Effects	YES	YES	YES	YES	YES	YES

Table 7. Real effects: Firms' ability to substitute loan supply decrease